Starfront Collaboration for N.I.N.A.

How to take part in a Starfront collaboration from N.I.N.A. with the Starfront plugin. For plugin 1.0.0.32, October 2026.

Before you start
Install the plugin
Sign in with Discord
Your filters and exposures
Join a collaboration
Start your own collaboration
7. Put it in your sequence
8. What happens during the night
9. Live stacking (optional)
10. Checking on it, and when something is wrong
11. Questions people ask

1. Before you start

You need N.I.N.A. 3.2 or later, membership of the Starfront Discord server, and a rig that can plate solve.

- **N.I.N.A. 3.2.0 or later.** Help > About shows the version. The plugin runs on 3.2 and 3.3 and will not install on older builds.
- **A Starfront Discord account.** Only members of the Starfront Discord server can sign in; there is no separate password or account, and you do not have to be a Starfront customer: anyone in the server, anywhere in the world, can join and start collaborations. You have to join it to take part: discord.com/invite/3ut6a7pfsY, also among the links at the top of the plugin's page under Plugins, and behind the **Join the Starfront Discord** button on its options page.
- **Plate solving that works.** Each field or panel is centred with N.I.N.A.'s own *Center* instruction, so your plate solver (ASTAP, PlateSolve2, Astrometry.net) must be set up under Options > Plate Solving and working. The *Center* instruction from the Advanced Sequencer palette is a good test.
- **Your usual profile, connected as usual.** Focal length (Options > Equipment > Telescope) and pixel size (camera) must be filled in; the server uses them to work out what your camera covers.
- **Your usual guider, autofocus and meridian-flip arrangements keep working**; the plugin only decides where to point and what to shoot. When a guider is connected, every field starts guiding after it is centred, the way a target container does.

2. Install the plugin

The plugin comes from Starfront's own plugin repository, not from the public list, so N.I.N.A. has to be told where to look once.

1. In N.I.N.A. go to **Options > General** and find **Plugin Repositories**.
2. Press **+** and enter exactly:

   ```
   https://collab.starfront.space/nina
   ```

3. Go to the **Plugins** tab (left sidebar) and open **Available**. **Starfront** is listed; press **Install**.
4. **Restart N.I.N.A.** Plugins only load at start-up.

5. Afterwards the plugin appears under Plugins > Installed, and updates show there too; install them like any other plugin update.

If Starfront is not listed, check the address for typing errors, then press the refresh button on the Plugins tab. The server address must be reachable from the PC; a browser visit to `https://collab.starfront.space/api/v1/health` should show a short line of text.

3. Sign in with Discord

Signing in once ties this PC's telescope to your Discord account; after that the plugin checks in with the server by itself.

1. Go to **Options > Plugins** and select **Starfront** in the list. Its page opens on the right.
2. Under **Account**, press **Sign in with Discord**. Your browser opens on Discord's authorisation page.
3. Sign in to Discord if asked, then **Authorize**. Discord only tells Starfront who you are and which server roles you hold; it cannot read your messages.
4. The browser shows **Signed in as <your name>**. Go back to N.I.N.A.: the page now says you are signed in and that this telescope is enrolled under your account. Enrolment happens by itself and names the telescope after your N.I.N.A. profile.
5. Signing in checks in with the server straight away, and the plugin keeps checking in on its own from then on. The status line should read **Checked in just now**.

If the page says the sign-in code expired, press the button again; you have ten minutes each time. If Discord says you are not a member of the server, join the Starfront Discord first (**Join the Starfront Discord** beside the sign-in button, or discord.com/invite/3ut6a7pfsY).

Several telescopes? Sign in on each PC. Each gets its own telescope under your account, and the server deals each one its own share. Signing in on a second PC may sign the first one's *account* out, but the telescope stays enrolled and keeps working; nothing needs redoing.

Sign out forgets the sign-in and the telescope on this PC. Frames already shot stay recorded locally and are reported after the next sign-in.

4. Your filters and exposures

You keep choosing how your frames are exposed; the collaboration only decides which filters tonight and how many frames go where.

On the same options page, under **Your exposures**, the table lists the filters of your N.I.N.A. filter wheel by their own names, with two things to set for each.

Delivers says what a collaboration gets from a sub through that filter. A project asks for channels, Ha, OIII, SII, red, green, blue and luminance, and for the colour-camera bands it is willing to take. A mono Ha delivers Ha. A one-shot colour camera's data is its own band, kept apart from the mono channels because a Bayer matrix's red is not a red filter's: its clear or UV/IR cut filter delivers **OSC-RGB**; an L-eXtreme, L-eNhance, L-Ultimate, ALP-T, NBZ or Askar D1 in front of it delivers **OSC-Ha+OIII**; an Askar D2 OSC-SII+OIII; a Triad Ultra or quad-band OSC-Ha+OIII+SII. The plugin guesses from the name and is usually right, so check the column once. The server then deals you the filters that deliver what a project takes, and a night through a dual-band filter is logged under its own band. A project that lists

mono channels only has nothing for a colour camera; its coordinator decides whether to take OSC broadband, OSC narrowband, both or neither.

Bortle class, under *This telescope*, is how dark your sky is, 1 (darkest) to 9. A collaboration may ask for a sky no worse than a given class; if you have not rated yours you are told what is wanted rather than turned away, and a night reported from a brighter sky than the project takes is rejected.

Seconds per sub is the length you normally shoot that filter at, for example 300 for Ha and 120 for luminance. Your share of every collaboration is dealt at these lengths, so every light calibrates against the darks you already have. The server works out for itself that your "Ha 3nm" is the project's Ha; you never rename anything.

A project may set a shortest or longest sub it will accept. If one of yours falls outside, joining tells you which filter to change, by its wheel name.

Under Camera

Setting	What it does	Usual value
Gain, Offset	Used for every collaboration frame	-1 = the camera's default from your profile
Binning	Binning of collaboration frames	1
Dither after every N frames	A dither trigger inside each field or panel	0, and keep using your own dither trigger around the container
Recentre after drifting (arcmin)	Each field checks its own drift after every frame and recentres on its own coordinates when it crosses this	5. Do not add a Center After Drift trigger above the container; see step 6
Rotate to the project's angle	With a rotator connected, turn the camera to the angle the project asks for	ticked
Camera angle (no rotator)	Read automatically from the first plate solve of each field	nothing to type

Under This telescope

Setting	What it does
Name on the chart	How your telescope is listed to everyone else; your profile's name unless you type one
Show where I point	Sends your mount's RA and Dec and what it is doing to the others' sky charts. Never your location
Bortle class of the sky	1 is the darkest sky, 9 an inner city; 0 is not rated. A collaboration may ask for a sky no worse than a given class

The server learns what your telescope can see from the camera itself: connect it once with the plugin installed, with the focal length set under Options > Equipment > Telescope. Until then, joining is refused with a line saying to connect your gear first.

How much of a night you give is not a setting: the sequence decides. The server deals a share sized to a full night, and the container shoots as much of it as the time you give it allows, in your own conditions and loop limits. The plugin checks in with the server every ten minutes on its own.

5. Join a collaboration

Joining is done from the **Starfront** panel on the Imaging tab, and it is a decision you make; nothing is ever shot for a project you have not joined.

1. Open the **Imaging** tab. In the panel list at the top (the row of icons where Image History and HFR History live) switch on **Starfront**. It docks like any other panel; drag it where you want it.
2. The panel shows the server status, who is online, and every open collaboration as one line: name, who started it, single field or mosaic and its size, the filters, how far along it is, hours collected, who is on it, how high it is in your sky right now, and a green or orange mark saying whether you can take part. Type a word in the search box to keep only the cards it appears on; **Show** narrows the list to the ones you can join or the ones you are on; **Sort** orders it by most active, least done, highest in your sky now, newest or name, with the ones you are on always first. Click a line to open its card.
3. The card shows:
   - what it is: a single field or a mosaic, its size, and the hours wanted per filter;
   - how far it has got, per channel: a bar for each of Ha, OIII, red and the rest with the hours collected against the goal, then the total collected and how much of the field is at full depth (on a mosaic, ten hours on one panel of fifteen is a fifteenth done);
   - who is on it and how many of them are online now;
   - in one green or orange line, whether **your telescope can take part** and through which filters, or why not: focal length, star size, a filter too wide, a sub too short;
   - a picture of its sky with the region outlined in yellow, and a graph of the target's altitude through tonight: dusk to dawn, your horizon and the project's floor shaded, transit marked, with one line saying when it is up. Once you have joined, your own panels are drawn on the sky in white and tonight's share in green. **Show in Framing Assistant** opens N.I.N.A.'s Framing Assistant on the region at the project's angle, with your camera's footprint.
4. Press **Join** on a collaboration you qualify for. The server hands your telescope its own part of the sky and, if it is already night, deals tonight's work straight away. The panel's **Tonight** line shows it, for example *M31: the field, 14 x Ha, 10 x OIII* or *Heart Nebula: 4 panels, 12 x OIII each*.
5. **Leave** takes you off it. Frames already shot stay credited to the project.

You can be on several collaborations at once; the sequence container shoots all of them, or the one you pick on it.

Sending your lights. Some collaborations collect the subs themselves for the coordinator to stack; the card's notes say so. Those go through the files page, collab.starfront.space/files: sign in with Discord there (it does not sign N.I.N.A. out), pick the collaboration, and drop the night's folder on it. Only the FITS files are taken, each filed as a light, flat, dark or bias by the word N.I.N.A. put in its name, and a file already there is skipped, so you can drop the same folder again the next morning. Send your calibration frames too: flats for every filter you shot, and darks and bias at the same gain,

offset, binning, temperature and exposure as your lights; the coordinator cannot calibrate your subs without them. A later version of the plugin will send each light as it is saved.

6. Start your own collaboration

Anybody signed in may start one; being in the Starfront Discord server is the whole gate. Open **Start a collaboration** at the top of the Starfront panel.

1. **Name** it. This is what everybody sees in their list.
2. **Where.** Type a name (*M31*, *NGC 7000*, *Sh2-129*) and press **Look up**; the server fills in the coordinates and, for the objects it knows, suggests a size. Or press **From Framing Assistant** to take whatever the Framing Assistant is showing, with its rotation. Or type RA and Dec yourself.
3. **Shape**: one field or a region to cover. One field means everybody points at the same spot and the depth map grows from the union of their cameras. A region is tiled by each telescope with its own camera; give its width and height in degrees.
4. **Filters wanted.** One row per filter, pre-filled from your own wheel. *Hours wanted* is the depth you want at every point of the field; blank means shoot it with no goal. *Bandpass* keeps wide filters out: *7* under Ha refuses a 12 nm Ha.
5. **Colour cameras.** Two boxes say whether the collaboration takes OSC broadband (OSC-RGB) and OSC narrowband (OSC-Ha+OIII). Ticking one puts that band among the filters wanted, with its own hours, kept apart from the mono channels in the depth map and the progress.
6. **Rules** are all optional: focal length and exposure ranges, the largest star size and guiding error a sub may have, how bright the Moon may be and how far away, the lowest altitude, the darkest Bortle class a contributing sky must have, the fewest frames a telescope is dealt on a panel in a night, calibrated frames only.
7. **Collect the lights on the server** if you mean to stack the data yourself. Everyone on it then sends their lights, flats, darks and bias through the files page, sorted by collaboration and person, and the card carries an *Open the files page* button. Off, the ledger records what was shot and the files stay with whoever shot them.
8. **Live stacking** needs nothing on this form: it follows the collaboration. If you run a live stacking hub, or a hub's owner has listed you as a coordinator on theirs, the server stacks the new collaboration there. Otherwise, if you have linked your Rapid Stacker Collab account (once, in the Collab tab of the Starfront program, under **Live stacking**), the server makes the Rapid Stacker Collab project itself and keeps its name, region, filters and rules in step every time you edit the collaboration. With neither, it is not stacked. To turn it off, or to choose another hub or the cloud, use the Collab tab of the Starfront program. There is no project to make on Rapid Stacker Collab and no join code to paste. A join code pasted into an older collaboration keeps working, and the form shows it when you edit that collaboration. Every telescope on the collaboration with Live stacking turned on sends its lights to the stack; see section 9.
9. **Start it.** It is live at once, in everybody's list. Join it yourself to shoot your share.

A collaboration you started shows **Edit**, **Deal again** and **Close** on its card. Edit changes anything, including the rules; nights already accepted stay accepted, and every telescope on it finds out at its next check-in. Deal again forgets tonight's lists so each telescope is dealt afresh when it next checks

in; use it after changing the filters or goals. Close takes it off everybody's list and keeps what was collected on record.

7. Put it in your sequence

One container does the whole night: drop **Starfront collaboration** where a target would go and keep your usual triggers around it. New to the Advanced Sequencer? The companion *Getting started with Starfront in N.I.N.A.* (collab.starfront.space/nina/start) builds this sequence one drag at a time.

1. Open the **Sequencer** tab in **Advanced** mode (the plugin cannot be used from the simple sequencer).
2. In the instruction palette on the right, find the **Starfront** category and drag **Starfront collaboration** into the **Targets** area of your sequence, where you would normally put a Deep Sky Object container.
3. On the container, **Collaboration** chooses what it shoots: *Every collaboration I have joined* (the default), or one project.
4. Put the container **directly in the Targets area**, never inside one of your own Deep Sky Object containers. A trigger on a target container takes its coordinates from *that* target: one tester's Center After Drift trigger, sitting on an M33 container around the Starfront one, measured 889 arcminutes of "drift" and slewed the mount to M33 in the middle of the M31 run. Nesting also costs a slew and a plate solve to the wrong target at every start.
5. Triggers that do not care about the target, *Autofocus after HFR increase* or *after time* and *Meridian Flip*, go on the Starfront container itself and fire around every frame. **Do not add Center After Drift or Dither after exposures above it.** Every field the container builds centres itself, dithers at the interval set in the options, and recentres on its own coordinates if it drifts further than the arcminutes set in the options. You will see that Center After Drift trigger inside each field while it runs; it is the plugin's own and should stay enabled.
6. Leave the container's own **Conditions** empty. It runs tonight's deal once and finishes; a *Loop until time* condition is honoured as a stop, never as a reason to shoot the deal twice.
7. Keep your **Start** and **End** areas as they are: cooling, unparking, guiding start, and parking and warming at the end.

The line under the container shows what the last check-in dealt, and the plugin flags anything missing before you run: not signed in, no exposures set, camera or mount not connected.

A simple arrangement that works:

```
Start
  Cool camera, Unpark, Start guiding
Targets
  Starfront collaboration     triggers: Meridian Flip, Autofocus after HFR increase
End
  Warm camera, Park
```

8. What happens during the night

When the sequence reaches the container it asks the server once what this telescope was dealt tonight, then works through it with N.I.N.A.'s own instructions.

The deal. The server has every telescope's part of the sky and everything anyone has already collected. Each night it deals you the frames to put on your field, or on a handful of panels of a mosaic, sized to the hours you said you give. On a mosaic that is one filter for the whole night, narrowband on a bright Moon, broadband or OIII on a dark one. On a single field it is one or more filters, starting with the one the project is furthest behind in against its goal, so the field's depth evens out across the filters; a night cut short loses frames from the deepest filter, not the thinnest. A night's deal holds for the whole night: nothing changes under a running sequence, however many frames arrive from elsewhere at 2 a.m. The next night is dealt afresh from what everyone really collected, and a project that adds a filter deals it to you on the next fresh deal if your wheel carries it.

The run. On a mosaic, if your camera has no rotator and the plugin does not yet know the angle it sits at, or the panels were laid for a different angle, the container first slews to the first panel and plate-solves once; the server then lays your panels again for the real angle before anything is shot. The container builds one target per field or panel, visible in the sequencer as it runs, each containing: *is it up?*, *Slew and center* (or *Slew, center and rotate* with a rotator), *Switch filter*, *Take many exposures*. Fields that are up go first, the one that sets soonest at the front; fields still to rise come after. The Starfront panel on the Imaging tab lists them all with where each has got to: done, shooting (frame 4 of 12, or slewing and centring), waiting and when it rises, skipped and why. The list updates with every frame and stays until the next run.

Stopping and starting. A sequence started again during the night carries on where it left off: frames already shot on a field tonight come off its count, and a finished field is not shot again.

Skipping. A field below the project's altitude floor, below your horizon file, or too near a Moon that is up is skipped for now, and the container comes back to it once the others are done. One that never clears the floor in the next twelve hours is left out of tonight and logged.

Your frames. Each light is saved by N.I.N.A. as usual. Its target name is the project's name, plus *panel 7* on a mosaic, so your file pattern's `$$TARGETNAME$$` sorts them. The FITS header carries `STARFRNT` (the project) and `SFPANEL`.

Reporting. Every light is credited to the sky it covered, with its star size and guiding error, kept on this PC, and sent to the server at the next check-in. If the server was down, the frames go the next time it answers; nothing is lost. The server judges each night against the project's rules and the panel shows accepted or rejected, with the reason.

When it ends. The container finishes when the deal is shot, and your End instructions run as normal. If you have time to spare after it, put another target after the container.

9. Live stacking (optional)

A collaboration that is live-stacked has everyone's lights stacked as they land: each light goes to its stacking project, which folds it into one deep stack per filter that grows through the night, for everybody on it to watch. The stacking project is either on the Rapid Stacker Collab cloud or on a live stacking hub, a computer its owner runs, often on the same home network as the telescopes. Either

way it is made and kept in step from the collaboration by the Starfront server, so there is no join code to find (section 6). The card says *Live-stacked on Rapid Stacker Collab*. Taking part is your choice, and is off until you turn it on.

1. Under **Options > Plugins > Starfront**, scroll to **Live stacking** and turn on **Send my lights to live stacking**.
2. For a collaboration stacked on the cloud, press **Sign in to Rapid Stacker Collab**. As with Discord, a browser window opens with a code to approve; there is no password to type into N.I.N.A. This account is separate from your Discord sign-in, and lasts about six months; the options page says how long it has left. A collaboration stacked on a hub needs no Rapid Stacker Collab sign-in: the Starfront server hands your telescope a pass to the hub at each check-in, and the plugin asks for the sign-in only when a cloud collaboration needs it.
3. That is all. At the next check-in the telescope joins the stacking project of every linked collaboration you are on. **Linked collaborations** on the options page lists them, each marked *Cloud* or *Hub:* and the hub's name, with how each of your filters is sent (from the *Delivers* column of your exposure table and the filter's name; a filter the project does not use is not sent). For a cloud project you can change that mapping on the Rapid Stacker Collab website.

What is sent. Only the lights the Starfront container saves for a linked collaboration, and only if they were pointed at the stacking project's field: a light pointed well away from it (another target, or a target name reused) could only be rejected there, so it stays here. Never your own file: the plugin makes a copy with your site's latitude, longitude and height, the observer's name, and the altitude and azimuth (which would give the site away) taken out, sends the copy and deletes it. Your files are never moved, changed or deleted.

Flats and darks. A flat goes to the stacking projects that got your lights of the same filter that night; one taken in the evening, before any light, waits for them until the next midday. Flats of a night you spent on another target stay here. Darks, bias and dark flats go to your telescope's **calibration library** on Rapid Stacker Collab, which calibrates its lights in every project, so they are sent once rather than to each project. Under **Calibration library** you can also choose a folder of them (masters or raw frames) to keep in step with the library; nothing is ever removed from it by the plugin. **Send flats** and **Send new darks there** turn either off; **Also send darks, bias** sends darks to the project of the last light sent instead, as before.

To a hub. On the hub's home network the plugin sends straight to the hub's own address; anywhere else, or whenever the hub does not answer there, the lights go through the Starfront server instead, and the log says each time it switches. If a hub refuses the telescope's pass, only that hub's lights wait, and they go with the fresh pass from the next check-in, without a restart. Cloud uploads carry on meanwhile.

When it is sent. By default *when the camera is idle*, between exposures, so an upload never competes with a download from the camera. You can choose *as soon as a frame is saved*, or *only inside a daily time window* (say 06:00 to 14:00, for a slow line), and set a speed limit. Uploads that are cut off, by a restart or a dropped connection, carry on where they stopped.

On the Starfront panel a **Live stacking** block under Tonight's deal shows how many frames are waiting and sent, the one going now, why uploads are holding if they are, and anything you need to do. **Pause uploads** holds everything until you resume; **Upload now** sends what is waiting whatever the mode says.

Terms. Joining a stacking project accepts Rapid Stacker Collab's terms of contribution as they stand, as the line beside the switch says; read them on its website before turning Live stacking on. If they change in the moment between the plugin reading them and joining, that join waits: press **Read the new terms** and **Accept them and carry on** under Linked collaborations.

> Live stacking sits beside the rest of the plugin, never in its way: the ledger, the reports and the night's deal are the same with it on or off, and with it off the plugin connects to nothing new.

10. Checking on it, and when something is wrong

Everything the plugin does is written to N.I.N.A.'s log with the prefix **Starfront:** (**Starfront stacking:** for live stacking), so the log answers most questions.

The log is under `%LOCALAPPDATA%\NINA\Logs` (paste that into the Windows Explorer address bar); the newest file is tonight's. Search it for `Starfront:`. The Starfront panel's status line and its **Frames reported** list show the same things in short.

What you see	What it means	What to do
Not signed in	No Discord sign-in on this PC	Options > Plugins > Starfront > Sign in with Discord
This telescope's token is no longer accepted	The telescope was removed on the server, or the settings file was copied from another PC	Sign out, then sign in again
the collaboration server could not be reached	No internet, or the server is down	The deal already in hand still runs; frames are reported when it answers
set an exposure for Ha in the exposure table first	A filter the project wants has no seconds against it	Fill it in under Options > Plugins
this telescope shoots Ha at 120s and the project wants 300s or longer	Your sub length is outside what the project accepts	Change that filter's seconds, or choose another project
Nothing dealt to this telescope tonight	You joined, but it is not night yet, or the collaboration is finished for your part of the sky	Check the Tonight line after dark
skipped for now: at 18°, below the 25° floor	The field is too low right now	Nothing; the container comes back to it
rejected - stars averaged 4.2", project wants 3.5" or better	The server judged the night against the project's rules	Nothing is lost locally; the coordinator can overrule a verdict
Plugin not in the Plugins list	N.I.N.A. older than 3.2, or the repository address mistyped	Update N.I.N.A.; check Options > General > Plugin Repositories
Meridian Flip ran at the very start of the container	Plugin older than 1.0.0.12	Update the plugin from Plugins > Installed
Sign in to Rapid Stacker Collab ... to start sending	Live stacking is on but has no sign-in, or its sign-in lapsed	Options > Plugins > Starfront > Sign in to Rapid Stacker Collab
The terms of M31 Deep changed; accept them to carry on	Rapid Stacker Collab's terms changed while the telescope was joining	Read and accept them under Linked collaborations; frames wait in the queue meanwhile
Joined M31 Deep; waiting for its owner to approve	The stacking project approves its members by hand	Nothing; frames wait and go once you are approved
Not joined: connect your camera first	The stacking project needs the camera's sensor and pixel size to describe your rig	Connect the camera once; it is tried again at the next check-in
Not joined: none of this wheel's filters is one the project wants	Your filters and the project's do not overlap, or the *Delivers* column says otherwise	Check the Delivers column of your exposure table, then **Retry now**

The plugin never touches your settings, profile or files beyond writing its own under `%LOCALAPPDATA%\NINA\PluginSettings` (live stacking keeps its queue and its temporary copies in a `stacking` folder there). Uninstalling it from the Plugins tab removes it completely.

11. Questions people ask

Can I still choose my exposure times? Yes; that is what the exposure table is. The collaboration decides which filters tonight and how many frames go where, never how a frame is exposed.

I set one hour but it dealt me more. That was a flaw in the server's dealing, fixed on 9 October 2026. A one-hour night now gets about an hour of one filter, the one the project needs most.

Will it move my rotator? Only if one is connected and *Rotate to the project's angle* is ticked. Without a rotator it shoots at the angle the camera sits and tells the server, so your panels are laid along your camera's axes.

What if clouds stop the night halfway? Whatever was shot is credited and reported. The next night's deal starts from what everyone really collected, so nothing has to be made up by hand.

Can I run my own targets the same night? Yes. Put them before or after the container in the Targets area, with their own conditions. The container takes the time your sequence gives it and no more.

Does it see my location? No. The server is told your focal length, pixel size, sensor size, filters, exposures and, if you allow it, where the mount points. Your site, horizon and local times stay on your PC; that is why the plugin works out the Moon and altitudes itself. With Live stacking on, the frames themselves go to Rapid Stacker Collab, with the site taken out of their headers first.

I have Starfront and N.I.N.A. on different telescopes. That is fine. Both talk to the same server, and the same person can own telescopes of either kind.

Who do I ask? The Starfront Discord server, where the people running the collaborations are. Say which telescope, and paste the `Starfront:` lines from the log for the night in question, or the whole log file.